

Exercise Sheet 2

System Safety

This sheet introduces key safety concepts and applies System-Theoretic Process Analysis (STPA) to the CARLA autonomous driving system described in the Course Roadmap. Refer back to that system description for all exercises below.

Hint: You are encouraged to solve these exercises digitally (e.g., using spreadsheets, digital diagrams, or notebooks). This will make it easier to later incorporate your results, diagrams, and evidence into your final safety report.

Terminology

Exercise 2.1: Safety Vocabulary

optional

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. **Loss** is an result of a situation that caused consequences like damages to humans, animals and objects or breaks of rules: a car crash
2. **Hazard** i a condition in witch a loss is likely to happen, the unsafe environment is uncertain of the outcome, missing a red light and driving through an intersection
3. **Risk** the likelihood and drastecality of a loss in a situation and the mesurement of repairability to revert the consequences, the associated cost of driving over an red intersection

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based *only* on the system description in the course Roadmap (you have not yet seen the training data), define the **Operational Design Domain** (ODD) for the CARLA system. Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed. ODD: sunny daytime, human safety operator, urban road, up to 50km/h, dry, mapped area

For each dimension, state: NODD: cloudy,rain onset, low sun angle, wet streets, sensor inop, no human operator,, long operator shift >4h, system malfunction, offroad, offmap, faster than 50km/h, reverse driving

1. The **operating conditions** (where the system is considered valid).
2. The **non-operating conditions** (where it is not).
3. How an ODD violation could be **detected at runtime**. 3. through human operator or bad illumination of sensor -> to dark/ to Bright > mean brightness out of range

Hint: You are defining the ODD *before* looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

Exercise 2.3: Losses

STPA begins by identifying the **losses** that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

ID	Loss Description	Why Unacceptable
L1	Loss of human life	long term psychological harm to driver and the relatives, loss of productive members of society,
L2	Injury to human pedestrians or drivers	health care costs of healing process, reduced productivity of member, long term pain and suffering
L3	Damage to road infrastructure and property	high repair and maintainance cost, legal costs, reduced trust in operation safety
L4	damage to the vehicle itself or other road vehicles	repair costs, legal costs, image drop, higher insurance costs

traffic law violations, public trust in a brand, health of the passenger of the car

Exercise 2.4: Hazards

A **hazard** is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the *system*, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

ID	Hazard	Loss(es)	Likelihood	Severity
H1	Pedestrian in critical distance in front of the car	L1, L2	medium	high
H2	vehicle misses to detect traffic light	L1,L2,L3,L4	low	high
H3	vehicle driving to fast/slow	L1,L2,L3,L4	medium	low
H4	vehicle driving to close to other road vehicles	L4,L2	high	low

h5 bad weather conditions, sensor malfunction human operator failing, intervene to late

STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
- **Control actions** (solid arrows): commands flowing from controllers to controlled processes (e.g., “brake command”).
- **Feedback** (dashed arrows): information flowing back (e.g., “perception labels”, “distance estimates”, “vehicle speed”).

Hint: There are at least two controllers in this system: the planning module and the human operator. The perception models and distance oracle should be considered *sensor/feedback*, not controllers themselves - they provide information to the controllers.

Todo: pfeile beschriften, vehicle motion

Hint: The traffic-light model only detects presence, not traffic light state. Consider what decisions the planner can (or cannot) safely make with this information.

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

Using your control structure, identify **unsafe control actions** for at least **two different controllers**. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., “H-1”, “H-2”).

For each controller, consider all four UCA types:

1. **Not provided** - a required action is missing.
2. **Provided unsafely** - the action is issued when it should not be.
3. **Wrong timing / order** - too early, too late, or out of sequence.
4. **Wrong duration** - applied too long or stopped too early.

1. missing action
2. unsafe action
3. wrong timing
4. applied to long/ too short

Fill in the table below (add rows as needed):

ID	Controller	Control action	UCA type	Hazard(s)	Unsafe scenario
UCA-1	human	human misses to stop	1: missing action	H1,H2,H3,H4	planning module is disengaged or ODD is left
UCA-2	human	human breaks without reason	2. provided unsafely	H3,H4	operator mistakenly gets on the break and deactivates the PM
UCA-3	human	Human steers or breaks panicly	3: wrong timing / order	H1,H2,H3,H4	PM deactivates abruptly in a difficult situation,
UCA-4	human	human breaks to little after deceleration	4. wrong duration	H2,H3,H4	HO is not seated correctly to break
UCA-5	planning module	break	1: missing action	H1,H2,H3,H4	ODD checks failed, human takeover missing
UCA-6	planning module	break	2. provided unsafely	H3,H4	over sensitive PM, low distances to other
UCA-7	planning module	break	3: wrong timing / order	H1,H2(driver);H3,H4	drivers behind emergency breaking is unsafely controlled->slip
UCA-8	planning module	break	4: wrong duration	H2,H3,H4	PM stops to break because critical velocity is left, but car might still crash or ditch

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one **safety constraint** - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- **Model-level** - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- **System-level** - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

UCA	Safety constraint	Level	Verification
1 2	human is required to use PM or steer / brake manually human control loop enforced.	2 protocol	OOD PM & Attention check
3	PM stays in control until HO takes it	2 fallback	fallback routine out of OOD to Driver -> auto decell
4	HO is required to stay in adequate seating position to take over in sub 2 seconds	2	takeover testing attention systems
5	PM OOD is restricted.	1	HO cover PM outside OOD OOC coverage of x%
6	emergency breaking must be assisted perception module should have low false positives rate, precision	1	low false negative rate
7	planning module know secure envelope to break -> not driving too fast on slippery ground or in corners	2	Break testings
8	standing animal !/ safe breaking	2	break to stop

STPA Step 4: Identify Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify **at least one causal scenario** - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. **The causal factor:** What component failure, misunderstanding, or adverse condition triggers the UCA?
2. **Why it occurs:** What is the root cause or underlying reason?
3. **Connection to constraint:** Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

Hint: Consider the following causal mechanisms (from the lecture):

- **Incorrect feedback:** A sensor or perception model provides wrong information to the controller.
- **Flawed internal model:** The controller misinterprets correct information (e.g., low confidence treated as “no detection”).
- **Timing delay:** Information arrives too late for the controller to respond safely.
- **Human factor:** The human operator fails to intervene due to attention lapse, delayed reaction, or mode confusion.

Fill in the table below (add rows as needed):

UCA	Causal scenario	Root cause	Related constraint
1	human is not aware of situation or driving	mode confusion human factor	attention requirement -> mode change attention inattention break system fallback
2	unassisted human touches steering wheel in assist mode and it deactivates unnoticed	human factor mode confusion	tolerance until override stiff steering and breaking mechanisms
3 4	PM detects nOOD to late and required human to late	timing delay	attention wake phase for OOD leaving -> detect difficult scenarios
5	driving in fog	flawed internal model incorrect feedback	OOD detection
6		flawed internal model	

contain a note:

traffic light detector -> planner has insufficient information for the status of the traffic light -> adapt system

Connecting STPA to the Rest of the Course

Exercise 2.9: Mapping Constraints to Evidence

optional

Review the safety constraints you derived in Exercise 2.7 and the causal scenarios from Exercise 2.8. For each **model-level** constraint, identify which future exercise sheet will provide evidence (refer to the Course Roadmap). For each **system-level** constraint, briefly describe what kind of evidence or argument you would need - even though you cannot run an experiment for it in this course.